

Juhi Patil

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Education

Purdue University, Elmore Family School of Electrical and Computer Engineering

West Lafayette, IN

Bachelor of Science in Electrical and Computer Engineering, Minor in Management

Expected Dec 2026

- Concentration: Artificial Intelligence & Machine Learning
- Relevant Coursework: Digital Signal Processing, Embedded Systems, Computer Security, Advanced C++ & Systems Programming

Experience

Midcontinent Independent System Operator (MISO)

Carmel, IN

Forecasting & Market Systems Analyst (Part-Time)

Aug 2025 – Dec 2025

- Maintained service-level documentation and standardized QA workflows for large-scale forecasting tools, improving knowledge transfer and reducing engineer onboarding time across cross-functional teams
- Built end-to-end technical information documentation for forecasting and cloud infrastructure, establishing reusable knowledge repositories that accelerated cross-team troubleshooting

Markets & Digital Strategy Intern

May 2025 – Aug 2025

- Elicited and consolidated requirements from 15-state SCADA telemetry sources; built a Python extraction and visualization framework that surfaced inverter irregularities and sub-MW renewable fluctuations previously undetected
- Managed end-to-end product change by migrating a deep learning model to Azure Machine Learning using Terraform, transitioning from local datasets to a cloud-scalable platform and reducing deployment time by 40%
- Awarded 1st Place, MISO Intern Data Challenge; recipient of 2025 Gulf Coast Power Association (GCPA) Scholarship

Seagate Technology

Longmont, CO

Firmware Engineering Intern

May 2024 – Aug 2024

- Managed a technical data repository of 500+ product requirements using an automated Excel-based validation system, reducing data errors by 50% and improving access efficiency by 35% to strengthen requirements traceability
- Resolved 50+ defects in test infrastructure via Perforce and Jira, applying structured troubleshooting and standardized documentation practices that cut suite maintenance time by 20%
- Led end-to-end test cycle analysis and communicated findings to stakeholders through structured Jira failure reports, reducing test cycle time by 25% and driving data-driven quality improvements

Projects

A.R.E.S. — Autonomous Robot Embedded System (Senior Capstone)

West Lafayette, IN

Embedded Software Developer · RP2350 / IAM-20380 / HC-SR04 / BLE

Jan 2025 – Present

- Developing embedded firmware for a two-MCU autonomous navigation platform; implementing I2C driver for IAM-20380 gyroscope, ultrasonic ranging, analog metal detector signal acquisition via ADC, and inter-MCU UART/BLE communication
- Writing and validating low-level peripheral drivers and integration test scripts in C, verifying sensor signal fidelity and timing correctness across the full embedded hardware stack

PurdueTHINK Consulting | The Data Mine Alumni Data Project

West Lafayette, IN

Project Manager

Aug 2024 – Dec 2024

- Elicited stakeholder requirements and designed an online data repository organizing 3,000+ alumni records by location, job title, and company, improving search efficiency by 40% and enabling targeted engagement tracking
- Developed a LinkedIn scraping tool and survey to collect and validate alumni data, increasing database coverage and boosting member retention and engagement by 20%

Leadership

PurdueTHINK Consulting

West Lafayette, IN

Consultant | Project Manager | Director of People

Dec 2024 – Present

- Drove end-to-end recruitment across 100–200 applications per cycle at a 10–15% acceptance rate, growing the applicant pool by 25% and boosting call-out attendance by 40%
- Planned and executed 20+ annual networking and team building events, overseeing logistics to improve member retention by 40%

Technical Skills

- **Software & Tools:** Python, SQL, JavaScript, C/C++, MATLAB, Microsoft Excel (VBA/Macros), Azure ML, Terraform
- **Project & Workflow Tools:** Jira, Perforce, Git, Agile workflows, QA process standardization, cross-functional coordination
- **Certifications:** Microsoft Office Specialist: Excel Associate | Python Programming Certification | NIMS CNC Mill Programming